

This is due by July 11

Problem 1. We show the Euclidean algorithm expresses the greatest common divisor of integers a, b as a integer linear combination of a and b . Let a, b be integers and $a > 0$. By Euclidean algorithm, we have the finitely many equations

$$\begin{aligned} b &= aq_0 + r_0 \\ a &= q_1r_0 + r_1 \\ r_0 &= q_2r_1 + r_2 \\ &\dots \\ r_{n-1} &= q_{n+1}r_n + r_{n+1} \end{aligned}$$

where $0 = r_{n+1} < r_n < \dots < r_1 < r_0 < a$.

1. Show for each $0 \leq k \leq n$, we can write $r_k = ax + by$ for some $x, y \in \mathbb{Z}$.
2. Use Euclidean algorithm to compute $d = (64, 28)$ and express $d = 64x + 28y$ for some $x, y \in \mathbb{Z}$
3. Prove that there are infinitely many integer solutions (x, y) to $d = 64x + 28y$.

Problem 2. Let $m > 0$. Suppose $a, b, c \in \mathbb{Z}$.

1. $a \equiv b \pmod{m}$ iff $m \mid a - b$
2. $a \equiv a \pmod{m}$ for all $a \in \mathbb{Z}$
3. If $a \equiv b \pmod{m}$, then $b \equiv a \pmod{m}$
4. If $a \equiv b \pmod{m}$ and then $b \equiv c \pmod{m}$, then $a \equiv c \pmod{m}$
5. Conclude congruence modulo m is an equivalence relation on $\mathbb{Z}$.

proof of part 4. Nikoli: Assume $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$. By part 3, $b \equiv a \pmod{m}$. Then (by Problem 3 part 3), $0 \equiv a - c \pmod{m}$ which implies $a \equiv c \pmod{m}$. $\square$

Problem 3. Problem 2 shows congruence relation is an equivalence relation on $\mathbb{Z}$. Let $m > 0$ and $a, b, a', b' \in \mathbb{Z}$ Define the congruence class $[a] = \{n \in \mathbb{Z} : n \equiv a \pmod{m}\}$. We call n an representative of $[a]$ for all $n \in [a]$. Define $\mathbb{Z}_m = \{[0], [1], \dots, [m-1]\}$.

1. Show that $a \equiv b \pmod{m}$ iff $[a] = [b]$.
2. Show that $a \not\equiv b \pmod{m}$ iff $[a] \cap [b] = \emptyset$.

3. Define functions $+$: $\mathbb{Z}_m \rightarrow \mathbb{Z}_m$ by $([a], [b]) \mapsto [a + b]$ and $\cdot$: $\mathbb{Z}_m \rightarrow \mathbb{Z}_m$ by $([a], [b]) \mapsto [ab]$. Show if $[a'] = [a]$ and $[b'] = [b]$, then $[a'] + [b'] = [a] + [b]$ and $[a'b'] = [ab]$.
4. Suppose $(a, m) = 1$. Then $\mathbb{Z}_m = \{[na] : 0 \leq n \leq m - 1\}$.

proof of part 2. It suffices to show $[a] \equiv [b]$ iff $[a] \cap [b] \neq \emptyset$ because part 2 is the negation of part 1. Then $[a] \cap [b] \neq \emptyset$ iff there exists some $x \in [a]$ and $x \in [b]$ iff $x \equiv a \pmod{m}$ and $x \equiv b \pmod{m}$ iff $a \equiv b \pmod{m}$. $\square$

proof of part 4. From part 3, we see $f : \mathbb{Z}_m \rightarrow \mathbb{Z}_m$ by $[n] \mapsto [n][a] = [na]$ is a well-defined map. Suppose $[i] \neq [j]$ for $0 \leq i < j \leq m - 1$ and $[ia] = [ja]$. Then $m \mid ia - ja = a(i - j)$. Since $(a, m) = 1$, by Euclid's lemma $m \mid i - j$. But $-m < i - j < m$ which implies $i - j = 0$ which contradicts $[i] \neq [j]$. So $[na] \subseteq \mathbb{Z}_m$ are all distinct so there are m such multiples in total. Yet $\mathbb{Z}_m$ has only m elements. Any subset of the same cardinality of a finite set must be the entire set. $\square$

Fred: We show f is injective. Suppose $[ia] = [ja]$ for $0 \leq i < j \leq m - 1$. Then $i = j$. So f is injective. Any injective map on a finite set is bijective.

Marcial: Since $(a, m) = 1$, then exists $xa + ym = 1$. In particular, the function $g : \mathbb{Z}_m \rightarrow \mathbb{Z}_m$ by $[a] \mapsto [xa]$ is the inverse function to f . That is, one may verify $fg = \text{id}_{[\mathbb{Z}_m]}$ and $gf = \text{id}_{\mathbb{Z}_m}$. So f and g are bijection and the image $\{[na] : 0 \leq n \leq m - 1\}$ is bijective to $\mathbb{Z}_m$.

Problem 4. We showed when a congruence equation has a solution in class. We now describe all the solutions. Let $r, m > 0$ be integers. Suppose $a, b \in \mathbb{Z}$.

1. Show that if $ra \equiv rb \pmod{m}$, then $a \equiv b \pmod{\frac{m}{(r, m)}}$
2. Show that if x_0 and x' are integer solutions to $ax \equiv b \pmod{m}$, then $x' \equiv x_0 \pmod{\frac{m}{(a, m)}}$
3. Find and explain all integer solutions to $2x \equiv 3 \pmod{5}$.

proof of part 4. Note that $2x \equiv 3 \pmod{5}$ implies $-x \equiv 4x \equiv 6 \pmod{5}$ which implies $x \equiv -5 \equiv 4 \pmod{5}$. Then it is necessary that $x \equiv 4 \pmod{5}$. Write $x = 4 + 5k$ for some $k \in \mathbb{Z}$. It is easy to verify that $2x \equiv 3 \pmod{5}$ always hold for any integer k (i.e, sufficiency is proved). So all the solutions must be of the form $4 + 5k$ where $k \in \mathbb{Z}$.

Marcial: $3 \cdot 2 \equiv 1 \pmod{5}$. So $2x \equiv 3 \pmod{5}$ is equivalent to $x \equiv 6x \equiv 9 \equiv 4 \pmod{5}$. In particular 3 is the multiplicative inverse to 2 when modulo 5. $\square$

Definition. A subset $I \subset \mathbb{Z}$ of the integers is called an ideal if

1. $0 \in I$,

2. for all $x, y \in I$, $x + y \in I$
3. for any $x \in I$, $nx \in I$ for all $n \in \mathbb{Z}$

Problem 5. Use the definition of an ideal in $\mathbb{Z}$ above. Let $a, b \in \mathbb{Z}$. Show the following.

1. Show if $I \subseteq \mathbb{Z}$ is an ideal, then there exists some $d \in I$ such that $d \mid r$ for all $r \in I$.
2. If $I, J \subseteq \mathbb{Z}$ are ideals, then $I \cap J$ is an ideal.
3. Show $(a) = \{na : n \in \mathbb{Z}\}$ is an ideal in $\mathbb{Z}$.
4. $(a) \subseteq (b)$ iff $b \mid a$
5. Show $(a) \cap (b) = (d)$ for some $d \in \mathbb{Z}$ and describe what is d .

proof of part 5. By part 2, we know $I = (a) \cap (b)$ is an ideal. By part 1 and 3, we know $I = (d)$ where d is the least positive number in $(a) \cap (b)$ unless $(a) = 0$ or $(b) = 0$. Then by part 4, $a \mid d$ and $b \mid d$. So d is the least common multiple of a and b . $\square$

Problem 6. Show the following.

1. Define $\tau(n)$ = number of positive divisors of n . Show that if $(m, n) = 1$ then $\tau(mn) = \tau(m)\tau(n)$.
2. Show there are infinitely many prime $\equiv 3 \pmod{4}$.
3. Is there an integer solution for the equation $x^3 - x + 1 = 0$? If yes, find all solutions. If no, prove it.

proof of part 1. Let $n = p_1^{e_1} \cdots p_n^{e_n}$ for primes $2 \leq p_1 < \cdots < p_n$ and $e_i \geq 0$ for $0 \leq i \leq n$. Then it is easy to show that $\tau(n) = (e_1 + 1) \cdots (e_n + 1)$.

Fred: Every positive divisor d of mn can be uniquely written as $d_1 \cdot d_2$ where $d_1 = (d, m)$ and $d_2 = (d, n)$ or $d/(d, m)$. Then $d \mapsto (d_1, d_2)$ is a bijection between $A = \{d > 0 : d \mid mn\}$ and $B \times C = \{d > 0 : d \mid m\} \times \{d > 0 : d \mid n\}$. $\square$

proof of part 2. Suppose there are only finitely many primes that is congruent to 3 mod 4. Say they are $P = \{p_1, \dots, p_n\}$. Then consider $N = 4p_1 \cdots p_n - 1$. Note N is either a prime or a product of primes. If N is a prime, then $N \in P$ which is a contradiction since $N > \max P$. Suppose $N = q_1 \cdots q_t$ for some primes q_i and $t \in \mathbb{N}_+$. Then we claim at least one of q_i for $1 \leq i \leq t$ is in P . Suppose not. That is, suppose each $q_i \equiv 1 \pmod{4}$ or $q_i = 2$. Since N is odd, $2 \nmid N$. So all $q_i \equiv 1 \pmod{4}$. Then $q_1 \cdots q_t \equiv 1 \pmod{4}$ and $N \equiv 3 \pmod{4}$. This is contradiction and we conclude at least one of q_i for $1 \leq i \leq t$ is in P . But none of $p_i \in P$ divides N for $N \equiv -1 \equiv 3 \pmod{p}$. So N is not composite nor prime, which is impossible. $\square$

proof of part 3. Suppose there is some solution x such that $x^3 - x + 1 = 0$. Consider $x \pmod 3$ and note $x^3 - x + 1 \equiv 0 \pmod 3$. Then $x = 0, 1, 2 \pmod 3$ which implies $x^3 = 0, 1, 8 = 0, 1, 2 \pmod 3$. In particular, $x^3 - x = 0 \pmod 3$. Then $x^3 - x + 1 \equiv 1 \pmod 3$ which is a contradiction.

Or one can use the rational root test.

Tianshuo: Note $x^3 - x = x(x-1)(x+1)$ which is a product of three consecutive integers. So it must be divisible by 3. Then $x^3 - x + 1 \equiv 1 \pmod 3$ which can't be 0 for all integer x .

Nikoli: Regardless of x being even or odd, $x^3 - x$ is even. Then $x^3 - x + 1$ is odd but 0 is even. $\square$